Non-Uniform Low-Light Face Image Enhancement Based on Dark Channel Prior & Image Uniform Posterior

A Project report submitted to
Jawaharlal Nehru Technological University, Kakinada
In the partial fulfillment for the award of the Degree of

BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the project entitled "Non-Uniform Low-Light Face Image Enhancement Based on Dark Channel Prior & Image Uniform Posterior" is *a bonafide work of*

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DECLARATION

We hereby declare that the work is being presented in this dissertation entitled "**Non-Uniform Low-Light Face Image Enhancement Based On Dark Channel Prior & Image Uniform Posterior**", submitted towards the partial fulfillment of requirements for the award of the degree of **Bachelor of Technology (ECE)** work carried out under the supervision of **K. Naga Hanumachari**, *Associate professor*, **RISE KRISHNA SAI GANDHI GROUP OF INSTITUTIONS**, Ongole. The results embodied in this dissertation report have not been submitted by me for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

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ABSTRACT

Non-uniform low-light images, characterized by complex lighting conditions, pose a significant challenge in image restoration and enhancement, particularly under backlit scenarios where the high-brightness background obscures the foreground information. Existing methods struggle with the diverse illumination factors across different scenes. To overcome this, we introduce a novel deep reconstruction network designed specifically for enhancing non-uniform low-light images. This network leverages the feature representation of uniform natural images, employing the dark channel of the absorption light scattering model (ALSM) to estimate and enhance facial features. Furthermore, the reconstruction process is optimized through the posterior constraint of uniform image characteristics, leading to superior detail enhancement. Our approach has been rigorously tested on both synthetic and real-world images, demonstrating its effectiveness in addressing the complexities of non-uniform low-light image enhancement. The results illustrate a notable enhancement, boosting PSNR by 28.37dB and SSIM by 0.0579 compared to recent methods, with a particular focus on enhancing facial feature details.

CHAPTER-1

INTRODUCTION TO IMAGE

1.1 INTRODUCTION

During the process of shooting, low lighting is often encountered due to environmental factors. Particularly in complex lighting conditions, the images can exhibit non uniform low-exposure states. For example, when facing the backlight source When catching a picture under faint light by utilizing a handheld camera, the inclination of the photographic artist's hand to shake makes the picture obscure. In light of this issue, picture de-obscuring has become a functioning point in computational photography and picture preparing as of late. From the perspective on signal preparing, picture de-obscuring can be diminished to a deconvolution issue if the piece capacity of the movement obscure is thought to be move invariant. However, the piece work isn't generally move invariant in genuine cases; for instance, in-plane revolution of a camera or a moving item can obscure various pieces of a picture as indicated by various part works. A picture that is corrupted by various haze bits is known as a non-uniform haze picture. In this paper, we propose a novel single picture de-obscuring calculation for non-uniform movement obscure pictures that is obscured by moving item. Initial, a proposed uniform defocus map technique is introduced for estimation of the sums and headings of movement obscure. These obscured areas are then used to appraise point spread capacities at the same time. a quick DE convolution calculation is utilized to re-establish the non-uniform haze picture. Dealing with complex light sources, or other similar situations, the critical foreground faces cannot be effectively identified, as depicted in Fig. 1(a). Existing illumination enhancement algorithms, such as those based on histogram correction or fusion techniques, find it challenging to achieve satisfactory enhancement in such scenarios.

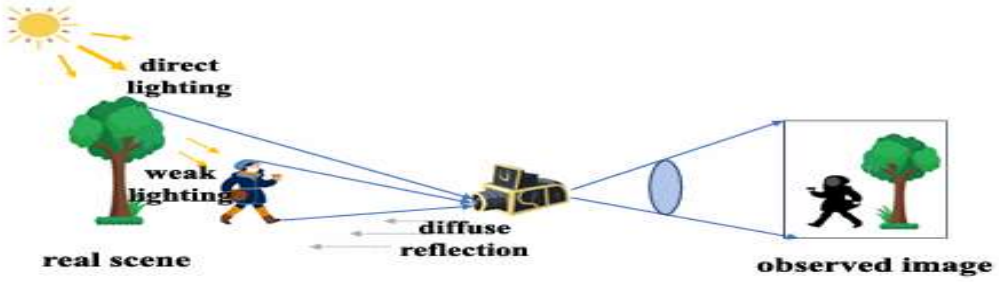


Fig 1.1: The scene of backlit images.

In the presence of direct sunlight, the background of the scene becomes overexposed while the foreground object appears underexposed due to glaring or diffuse reflections. Consequently, these complex lighting conditions pose challenges in capturing important information in the foreground of backlit images.

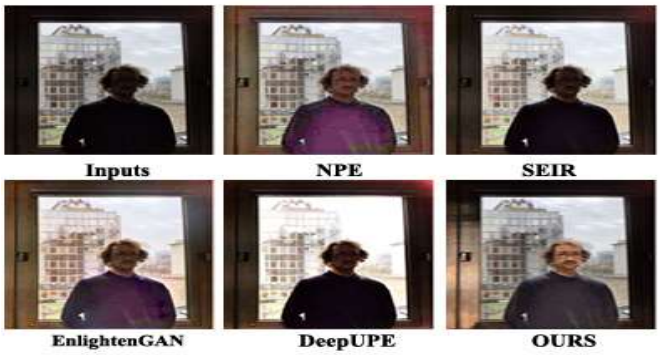


Fig 1.2: The results of both the compared methods

Our proposed method for a real backlit image are presented. Among these methods, our approach demonstrates superior performance in enhancing the details of the foreground faces.

1.2 IMAGE ENHANCEMENT

Image Enhancement is one of the most important and difficult techniques in image research. The aim of image enhancement is to improve the visual appearance of an image, or to provide a "better transform representation for future automated image processing. Many images like medical images, satellite images, aerial images and even real-life photographs suffer from poor contrast and noise. It is necessary to enhance the contrast and remove the noise to increase image quality. One of the most important stages in medical images detection and analysis is Image Enhancement techniques which improves the quality (clarity) of images for human viewing, removing blurring and noise, increasing contrast, and revealing details are examples of enhancement operations.

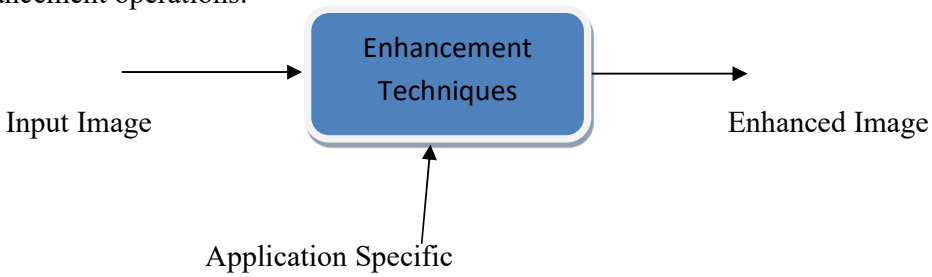


Fig 1.3: Basic Block Diagram of Image Enhancement

A. Scope

The scope of this project focuses on the development and evaluation of an image enhancement system specifically designed to improve facial images captured under **non-uniform low-light conditions**. Such conditions commonly occur due to uneven illumination, shadows, backlighting, or multiple light sources, which significantly degrade facial visibility and recognition performance.

B. Objective

The primary objective of this project is to develop an effective image enhancement framework that improves the visual quality and clarity of face images captured under **non-uniform low-light conditions**. The project aims to address uneven illumination, noise, and loss of facial details while preserving natural appearance and identity-specific features.

1.3 MOTIVATION

With the quick development of advancements and the commonness of imaging gadgets, billions of computerized pictures are being made each day. Because of unwanted light source, ominous climate or disappointment of the imaging gadget itself, the difference and tone of the caught picture may not generally be palatable. Actually, picture upgrade calculations have just been generally connected in imaging gadgets for tone mapping.

CHAPTER 2
IMAGE PROCESSING

2.1 DIGITAL IMAGE PROCESSING

A picture might be characterized as a two-dimensional capacity, f (x, y), where x and y are spatial (plane) organizes, and the adequacy of f at any pair of directions (x, y) is known as the power or dim degree of the picture by then. Whenever x, y, and the plentifulness estimations off are for the most part limited, discrete amounts, we consider the picture a computerized picture. The field of computerized picture preparing alludes to handling advanced pictures by methods for an advanced PC. Note that an advanced picture is made out of a limited number of components, every one of which has a specific area and worth. These components are alluded to as picture components, picture components, peels, and pixels. Pixel is the term most broadly used to signify the components of an advanced picture.

Vision is the most exceptional of our faculties, so it isn't astounding that pictures play the absolute most significant job in human observation. Be that as it may, in contrast to people, who are constrained to the visual band of the electromagnetic (EM) range, imaging machines spread nearly the whole EM range, going from gamma to radio waves. They can work on pictures created by sources that people are not familiar with partner with pictures. These incorporate ultra-sound, electron microscopy, and PC created pictures. In this manner, advanced picture handling incorporates a wide and shifted field of utilizations. There is no broad understanding among creators with respect to where picture handling stops and other related regions, for example, picture investigation and PC vision, begin. Now and again a qualification is made by characterizing picture handling as an order in which both the info and yield of a procedure are pictures.

We accept this to be a restricting and fairly counterfeit limit. For instance, under this definition, even the insignificant undertaking of figuring the normal force of a picture (which yields a solitary number) would not be viewed as a picture preparing task. Then again, there are fields, for example, PC vision whose extreme objective is to utilize PCs to imitate human vision, including learning and having the option to make inductions and take activities dependent on visual data sources. This territory itself is a part of man-made reasoning (AI) whose goal is to copy human insight.

The field of AI is in its most punctual phases of early stages as far as advancement, with advancement having been much slower than initially foreseen. The zone of picture investigation (additionally called picture understanding) is in the middle of picture preparing and PC vision. There are no obvious limits in the continuum from picture preparing toward one side to PC vision at the other. In any case, one valuable worldview is to think about three kinds of electronic procedures in this continuum: low-, mid-, and abnormal state forms. Low-level procedures include crude activities, for example, picture pre-processing to decrease clamour, differentiate upgrade, and picture.

2.2 DEFINITION OF IMAGE

An image is an array, or a matrix, of square pixels (picture elements) arranged in columns and rows.

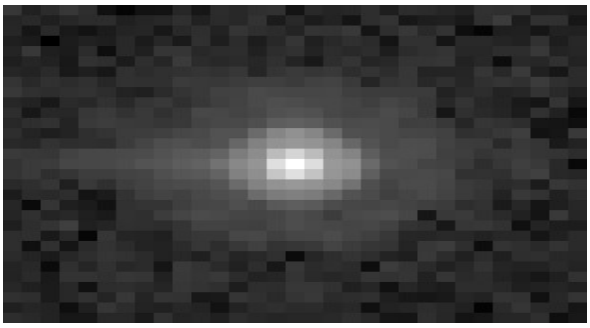


Figure 2.1: An array or a matrix of pixels arranged in columns and rows

In a (8-bit) greyscale picture each image component has a doled-out power that reaches from 0 to 255. A dim scale picture is the thing that individuals typically call a high contrast picture, however the name underlines that such a picture will likewise incorporate numerous shades of dim.

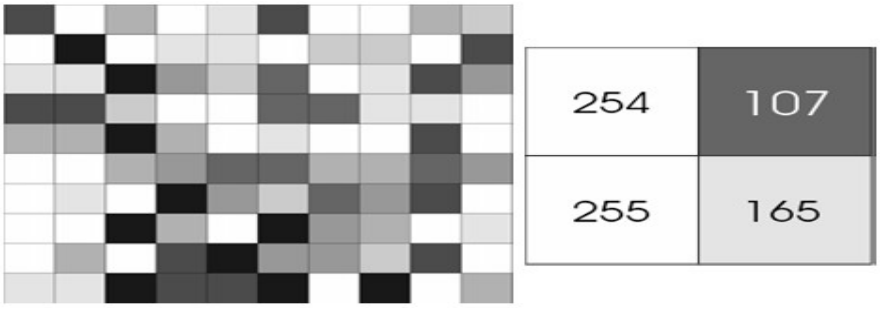


Figure 2.2: 8 bit = 256 tonnes or grayscales image

In Figure 2.2 Each pixel has an incentive from 0 (dark) to 255 (white). The conceivable scope of the pixel esteems relies upon the shading profundity of the picture, here 8 bit = 256 tonnes or grayscales. An ordinary grayscale picture has 8-bit shading profundity = 256 grayscales. A "real nature" picture has 24-bit shading profundity = 8 x 8 x 8 bits = 256 x 256 x 256 hues = ~16 million hues.

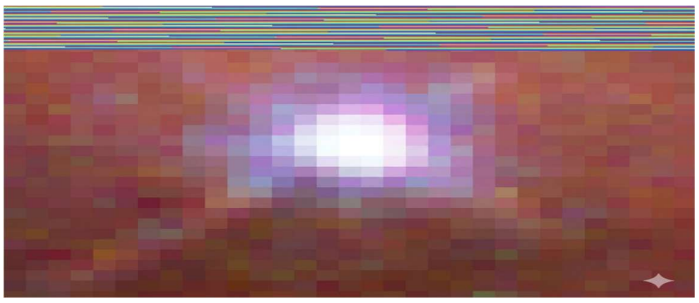


Figure.2.3: A true-colour image assembled from three grayscale images

Figure 2.3 demonstrates A genuine nature picture collected from three grayscale pictures hued red, green and blue. Such a picture may contain up to 16 million unique hues. Some grayscale pictures have more grayscales, for example 16 bit = 65536 grayscales. On a basic level three grayscale pictures can be joined to shape a picture with 281,474,976,710,656 grayscales. There are two general gatherings of 'pictures': vector illustrations (or line workmanship) and bitmaps (pixel-based or 'pictures').

Probably the most well-known document configurations are:

GIF

An 8-bit (256 shading), non-ruinously compacted bitmap design. For the most part utilized for web. Have a few sub-models one of which is the energized GIF.

JPEG

An exceptionally productive (for example much data per byte) ruinously packed 24-bit (16 million hues) bitmap group. Generally utilized, particularly for web and Internet (transmission capacity constrained).

TIFF

The standard 24-bit distribution bitmap format. Compresses non-damagingly with, for example, Lempel-Ziv-Welch (LZW) pressure.

PS

Postscript, a standard vector design. Has various sub-benchmarks and can be hard to ship crosswise over stages and working frameworks.

Pictures are the most widely recognized and helpful methods for passing on or transmitting data. Words generally can't do a picture justice. Pictures compactly pass on data about positions, sizes and entomb connections between articles.

They depict spatial data that we can perceive as items. Individuals are great at getting data from such pictures, as a result of our natural visual and mental capacities. About 75% of the data gotten by human is in pictorial structure. A picture is digitized to change over it to a structure which can be put away in a PC's memory or on some type of capacity media, for example, a hard circle or CD-ROM. This digitization methodology should be possible by a scanner, or by a camcorder associated with a casing grabber board in a PC. When the picture has been digitized, it tends to be worked upon by different picture preparing tasks.

Picture handling tasks can be generally separated into three noteworthy classes, Image Compression, Image Enhancement and Restoration, and Measurement Extraction. It includes decreasing the measure of memory expected to store an advanced picture. Picture absconds which could be brought about by the digitization procedure or by shortcomings in the imaging set-up (for instance, terrible lighting) can be redressed utilizing Image Enhancement strategies. When the picture is in great condition, the Measurement Extraction tasks can be utilized to get valuable data from the picture. A few instances of Image Enhancement and Measurement Extraction are given underneath. The models demonstrated all work on 256 dark scale pictures. This implies every pixel in the picture is put away as a number between 0 to 255, where 0 speaks to a dark pixel, 255 speaks to a white pixel and qualities in the middle of speak to shades of dim. These activities can be stretched out to work on shading pictures. The models underneath speak to just a couple of the numerous procedures accessible for working on pictures. Insights regarding the internal functions of the tasks have not been given, yet a few references to books containing this data are given toward the end for the intrigued per user.

2.3 IMAGES AND PICTURES

As we referenced in the prelude, people are prevalently visual animals: we depend intensely on our vision to comprehend our general surroundings. We not just see things to recognize and group them, however we can filter for contrasts, and acquire a general harsh inclination for a scene with a speedy look. People have developed exact visual aptitudes: we can recognize a

face in a moment; we can separate hues; we can process a lot of visual data in all respects rapidly.

Be that as it may, the world is in consistent movement: gaze at something for a considerable length of time and it will change somehow or another. Indeed, even an enormous strong structure, similar to a structure or a mountain, will change its appearance relying upon the season of (day or night); measure of daylight (clear or overcast), or different shadows falling upon it. We are worried about single pictures: depictions, in the event that you like, of a visual scene. In spite of the fact that picture preparing can manage evolving scenes, we will not talk about it in any detail in this content. For our motivations, a picture is a solitary picture which speaks to something.

It might be an image of an individual, of individuals or creatures, or of an open-air scene, or a microphotograph of an electronic part, or the consequence of restorative imaging. Regardless of whether the image isn't quickly conspicuous, it won't be only an irregular haze.

Picture handling includes changing the idea of a picture so as to either

1. Improve its pictorial data for human translation,
2. Render it progressively reasonable for self-sufficient machine observation.

We will be worried about computerized picture preparing, which includes utilizing a PC to change the idea of an advanced picture. It is important to understand that these two viewpoints speak to two isolated yet similarly significant parts of picture handling. A method which fulfils condition, a strategy which makes a picture look better might be the most exceedingly awful system for fulfilling condition. People like their pictures to be sharp, clear and itemized; machines incline toward their pictures to be straightforward and uncluttered.

2.4 IMAGES AND DIGITAL IMAGES

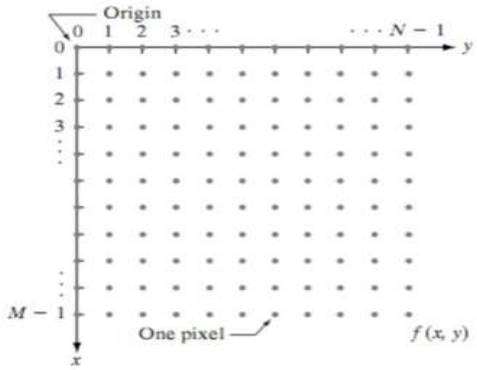


Figure 2.4: Coordinate convention used to represent digital mage

Assume we take a picture, a photograph, say. For the occasion, let’s make things simple and assume the photograph is high contrast (that is, loads of shades of dim), so no shading. We may

think about this picture just like a two-dimensional capacity, where the capacity esteems give the brilliance of the picture at some random point.

We may expect that in such a picture brilliance esteems can be any genuine numbers in the range (dark) (white). An advanced picture can be considered as an enormous exhibit of discrete specks, every one of which has a splendour related with it. These spots are called picture components, or all the more essentially pixels. The pixels encompassing a given pixel establish its neighbourhood. An area can be described by its shape similarly as a network: we can talk about an area. With the exception of in exceptionally extraordinary conditions, neighbourhoods have odd quantities of lines and sections; this guarantees the present pixel is in the focal point of the area.

2.5 FUNDAMENTAL STEPS IN DIGITAL IMAGE PROCESSING

Picture procurement is the primary procedure appeared in Fig.2. Note that obtaining could be as straightforward as being given a picture that is as of now in advanced structure. For the most part, the picture securing stage includes pre-processing, for example, scaling. Picture improvement is among the least difficult and most engaging regions of advanced picture handling.

Fundamentally, the thought behind improvement systems is to bring out detail that is clouded, or basically to feature certain highlights of enthusiasm for a picture. A commonplace case of upgrade is the point at which we increment the complexity of a picture since "it looks better." It is critical to remember that improvement is an emotional zone of picture preparing. Picture rebuilding is a zone that likewise manages improving the presence of a picture. Be that as it may, in contrast to improvement, which is emotional, picture rebuilding is objective, as in reclamation strategies will in general be founded on numerical or probabilistic models of picture debasement.

Upgrade, then again, depends on human emotional inclinations with respect to what establishes a "decent" improvement result. Shading picture preparing is a territory that has been picking up in significance in light of the critical increment in the utilization of advanced pictures over the Internet.

The pixels encompassing a given pixel comprise its neighbourhood. An area can be portrayed by its shape similarly as a network: we can talk about an area. With the exception of in extremely unique conditions, neighbourhoods have odd quantities of lines and segments; this guarantees the present pixel is in the focal point of the neighbourhood. Generally, the picture securing stage includes pre-processing, for example, scaling. Picture improvement is among

the least difficult and most engaging zones of advanced picture processing. We may expect that in such a picture splendour esteems can be any genuine numbers in the range (dark) (white). A computerized picture can be considered as a huge exhibit of discrete spots, every one of which has a brilliance related with it.

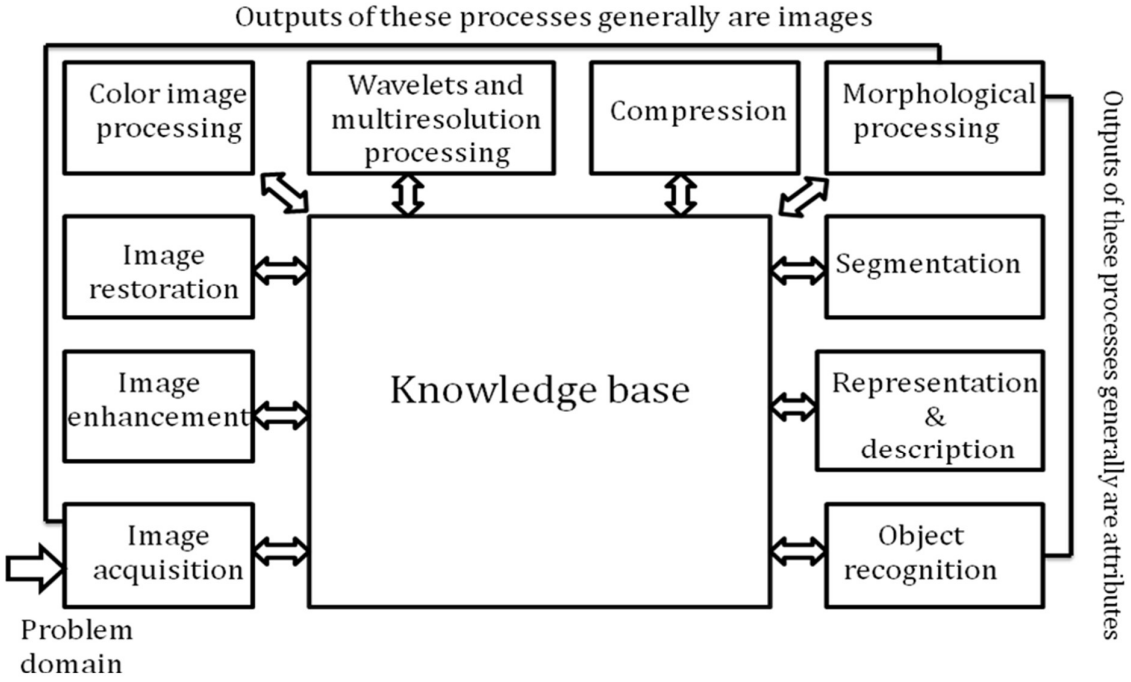


Figure 2.5: Fundamental steps in Digital Image Processing

Wavelets are the establishment for speaking to pictures in different degrees of goals. Pressure, as the name infers, manages systems for lessening the capacity required to spare a picture, or the transmission capacity required to transmit it. Despite the fact that capacity innovation has improved fundamentally over the previous decade, the equivalent can't be said for transmission limit. This is genuine especially in employments of the Internet, which are described by noteworthy pictorial substance. Picture pressure is natural (perhaps inadvertently) to most clients of PCs as picture record augmentations, for example, the jpg document expansion utilized in the JPEG (Joint Photographic Experts Group) picture pressure standard.

Morphological handling manages devices for removing picture segments that are helpful in the portrayal and depiction of shape. Division techniques parcel a picture into its constituent parts or articles. By and large, self-governing division is a standout amongst the most troublesome errands in computerized picture preparing. A rough division methodology brings the procedure far toward fruitful arrangement of imaging issues that expect articles to be recognized independently. Then again, feeble or unpredictable division calculations quite often ensure possible disappointment.

Portrayal and depiction quite often pursue the yield of a division arrange, which for the most part is crude pixel information, establishing either the limit of a district (i.e., the arrangement of pixels isolating one picture area from another) or every one of the focuses in the locale itself. In either case, changing over the information to a structure appropriate for PC preparing is vital. The principal choice that must be made is whether the information ought to be spoken to as a limit or as a total area. Limit portrayal is fitting when the emphasis is on outside shape attributes, for example, corners and articulations. Local portrayal is fitting when the attention is on inner properties, for example, surface or skeletal shape. In certain applications, these portrayals supplement one another. Picking a portrayal is just piece of the answer for changing crude information into a structure appropriate for resulting PC preparing.

A strategy should likewise be indicated for portraying the information so highlights of intrigue are featured. Depiction, likewise called highlight choice, manages separating traits that outcome in some quantitative data of intrigue or are essential for separating one class of articles from another. Acknowledgment is the procedure that doles out a name (e.g., "vehicle") to an item dependent on its descriptors. We finish up our inclusion of advanced picture preparing with the improvement of techniques for acknowledgment of individual items.

2.6 APPLICATIONS AND USAGE

Since computerized picture preparing has exceptionally wide applications and practically the majority of the specialized fields are affected by DIP, we will simply talk about a portion of the significant uses of DIP. Advanced Image preparing isn't simply restricted to change the spatial goals of the regular pictures caught by the camera. It isn't simply restricted to build the brilliance of the photograph, etc. Maybe it is undeniably increasingly over that. Electromagnetic waves can be thought of as stream of particles, where every molecule is moving with the speed of light. Every molecule contains a heap of vitality. This heap of vitality is known as a photon. The electromagnetic range as indicated by the vitality of photon is demonstrated as follows. In this electromagnetic range, we are just ready to see the noticeable range.

Noticeable range predominantly incorporates seven distinct hues that are normally term as VIBGOYR. VIBGOYR represents violet, indigo, blue, green, orange, yellow and Red. However, that does not invalidate the presence of other stuff in the range.

Our human eye can just observe the unmistakable part, in which we saw every one of the items. Be that as it may, a camera can see different things that awaked eye can't see. For instance: x beams, gamma beams, etc. Consequently, the examination of all that is tuff also is

done in computerized picture handling. The appropriate response lies in the reality, since that other stuff, for example, X-beam has been broadly utilized in the field of medicinal. The examination of Gamma beam is fundamental since it is utilized broadly in atomic medication.

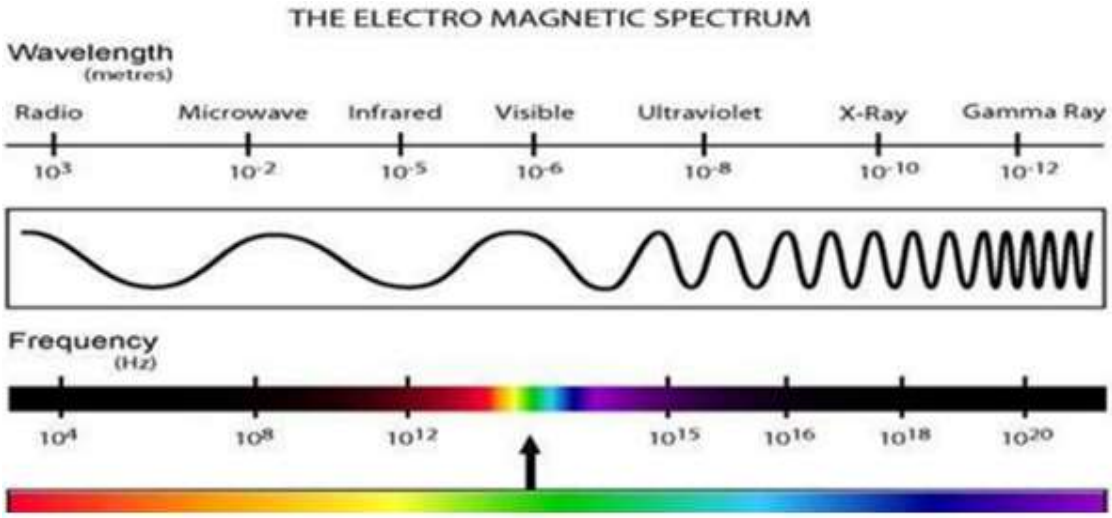


Figure 2.6: Electro Magnetic Spectrum

2.6.1 APPLICATIONS OF DIGITAL IMAGE PROCESSING

Some of the major fields in which digital image processing is widely used are mentioned below

1. Image sharpening and restoration
2. Medical field
3. Remote sensing
4. Transmission and encoding
5. Machine/Robot vision
6. Colour processing
7. Pattern recognition
8. Video processing
9. Microscopic Imaging
10. Others

2.7 IMAGE SHARPENING AND RESTORATION

Picture honing and reclamation alludes here to process pictures that have been caught from the cutting-edge camera to make them a superior picture or to control those pictures in approach to accomplish wanted outcome.

It alludes to do what Photoshop normally does. This incorporates Zooming, obscuring, honing dim scale to shading transformation, recognizing edges and the other way around, Image recovery and Image acknowledgment. Our human eye can just observe the noticeable bit, in which we saw every one of the articles. In any case, a camera can see different things that awaked eye can't see.

The common examples are:

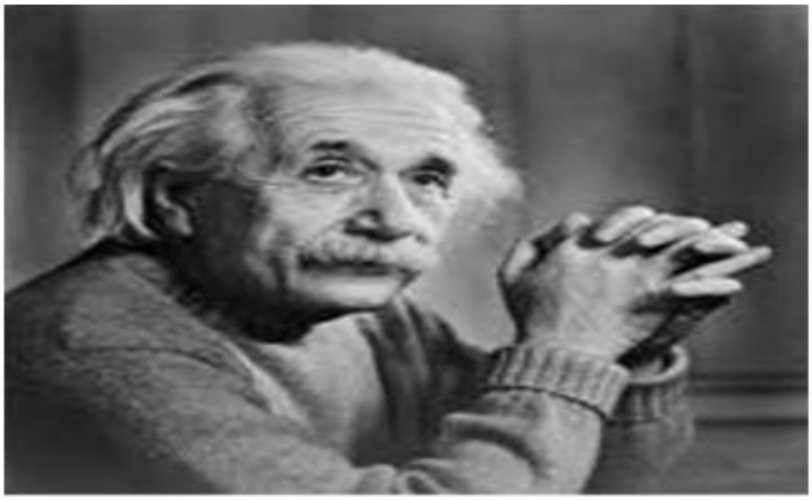


Figure 2.7: Original Image

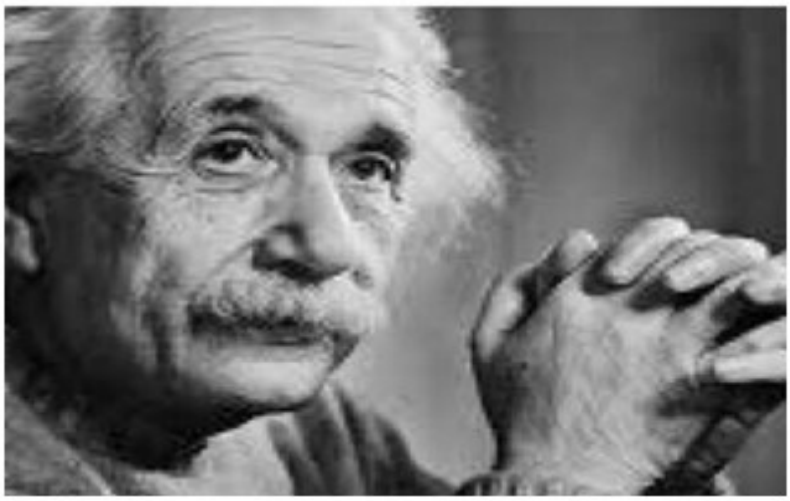


Figure 2.8: Zoomed Image

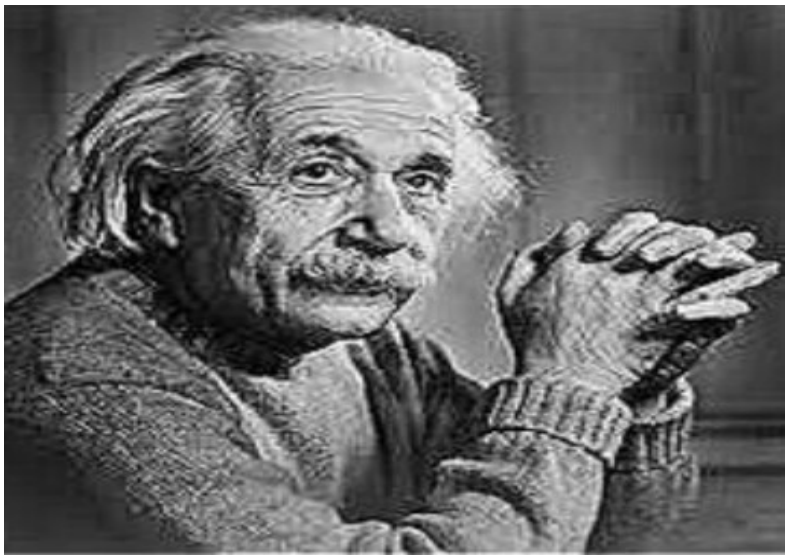


Figure 2.9: Sharp Image

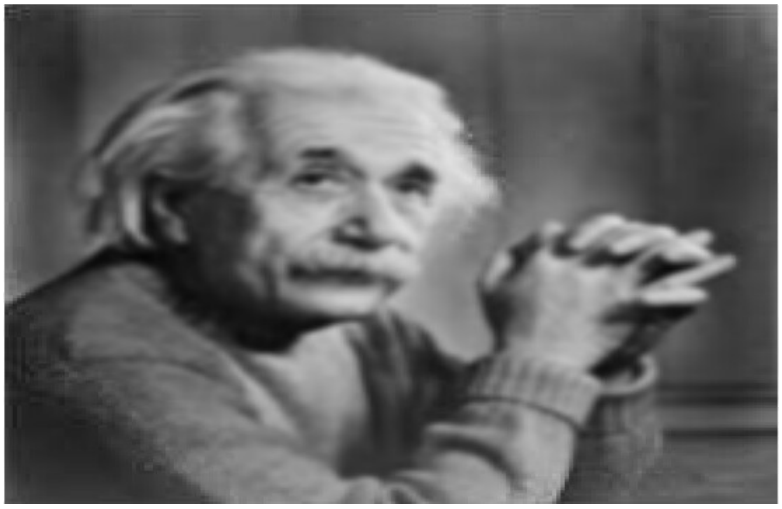


Figure 2.10: Blur Image



Figure 2.11: Edges

2.8 MEDICAL FIELD

The common applications of DIP in the field of medical is

1. Gamma ray imaging
2. PET scan
3. X Ray Imaging
4. Medical CT
5. UV imaging

2.8.1 UV imaging

In the field of remote detecting, the territory of the earth is filtered by a satellite or from an extremely high ground and after that it is investigated to get data about it. One specific utilization of computerized picture preparing in the field of remote detecting is to distinguish foundation harms brought about by an earth shake. As it requires longer investment to handle harm, regardless of whether genuine harms are centred around.

Since the region affected by the seismic tremor is once in a while so wide, that it impractical to analyse it with human eye so as to gauge harms. Regardless of whether it is, at that point it is extremely wild and tedious technique. So, an answer for this is found in computerized picture handling. A picture of the influenced territory is caught from the over the ground and afterward it is breaking down to identify the different sorts of harm done by the seismic tremor.

1. The key advances incorporate into the examination are
2. The extraction of edges
3. Analysis and improvement of different sorts of edges

2.9 VIDEO PROCESSING

A video is only simply the extremely quick development of pictures. The nature of the video relies upon the quantity of edges/pictures every moment and the nature of each edge being utilized. Video preparing includes clamour decrease, detail upgrade, movement recognition, edge rate change, angle proportion transformation, shading space transformation etc. The nature of the video relies upon the quantity of casings/pictures every moment and the nature of each edge being utilized.

The vast majority of the robots today work by following the line and in this way are called line devotee robots. This causes a robot to proceed onward its way and play out certain errands. This has additionally been accomplished through picture handling.

Example acknowledgment includes ponder from picture handling and from different fields that incorporates AI a branch of artificial knowledge. In example acknowledgment, picture

handling is utilized for distinguishing the articles in a picture and afterward AI is utilized to prepare the framework for the adjustment in example. Example acknowledgment is utilized in PC supported finding, acknowledgment of penmanship, acknowledgment of pictures etc.

Aside from the numerous difficulties that a robot face today, one of the greatest tests still is to build the vision of the robot. Make robot ready to see things, distinguish them, and recognize the obstacles etc. Much work has been contributed by this field and a total other field of PC vision has been acquainted with work on it. Colour preparing incorporates handling of hued pictures and diverse shading spaces that are utilized.

For instance, RGB shading model, YCbCr, HSV. It likewise includes considering transmission, stockpiling, and encoding of these shading images. In design acknowledgment, picture handling is utilized for recognizing the items in a picture and afterward AI is utilized to prepare the framework for the adjustment in example. Example acknowledgment is utilized in PC supported conclusion, acknowledgment of penmanship.

CHAPTER-3

LITERATURE SURVEY

3.1 PRIOR RESEARCH

[1]: Unsupervised learning non-uniform face enhancement under physics-guided model of illumination decoupling. This recent work addresses the specific challenge of enhancing non-uniform low-light face images without paired ground-truth data. The authors introduce a physics-guided illumination decoupling model that iteratively enhances images by separating illumination effects and updating face structure in a two-phase process. A custom non-uniform low-light face dataset (NUL-Face) is also presented. Experimental results show improved naturalness and visual quality compared with recent unsupervised methods.

[2]: Self-Supervised Non-Uniform Low-Light Image Enhancement Combining Image Inversion and Exposure Fusion. This paper proposes a self-supervised framework for enhancing non-uniformly illuminated low-light images using a Retinex-based exposure enhancement network (EENet) and a three-branch asymmetric exposure fusion network (TAFNet)**.** It eliminates the need for paired data and addresses over-exposure while recovering details and reducing noise through fusion of enhanced and inverted images.

[3]: Non-Uniform-Illumination Image Enhancement Algorithm Based on Retinex Theory Focusing on general non-uniform illumination (not just faces), this Retinex approach divides lighting into high and low areas, applies adaptive gamma correction for darker regions, and uses Weber–Fechner law for bright regions. A multi-scale detail enhancement step further improves visual quality, demonstrating clear improvement in non-uniform low-light conditions.

[4]: Local Content-Aware Enhancement for Low-Light Images with Non-Uniform Illumination. This model improves upon classical Retinex by leveraging local content awareness and multi-scale guided filtering to better manage uneven lighting and avoid common artifacts like halo effects. The approach is particularly effective at preserving edge details and content under strongly varying light conditions.

[5]: Non-uniform Low-Light Image Enhancement via Noise-Aware Decomposition and Adaptive Correction. This work introduces a noise-aware decomposition network that separates illumination, reflection, and noise components. Adaptive correction maps are then learned to suppress noise and correct brightness. Though not face-specific, the method is relevant for non-uniform low-light image pre-processing in face applications.

[6]: Non-Uniform Low-Light Stereo Image Enhancement via Transformer. This paper (from ACM DL) proposes a Transformer-based pipeline for non-uniform low-light enhancement in stereo images, integrating lighting adjustment and detail recovery stages. While focused on stereo setups, its Transformer strategy for handling uneven illumination can inspire face enhancement architectures.

[7]: Non-Uniform Low-Light Face Image Enhancement Based on Dark Channel Prior and Image Uniform Posterior. This research applies a dark channel prior and uniform posterior distribution to enhance low-light face images by restoring contrast and reducing uneven illumination. The reported results show significant improvement in PSNR and SSIM compared with existing enhancement Methods, Indicating Its Practical Value.

3.2 COMPARSION ANALYSIS

Ref No.	Year	Technique	PSNR (dB)	SSIM	Entropy	Accuracy
[1]	2022	Physics-guided Illumination Decoupling	22.04*	0.812*	—	High
[2]	2023	EENet + TAFNet (Exposure Fusion)	21.36	0.825	—	—
[3]	2024	Retinex + Adaptive Gamma Correction	20.84	0.852	7.643	—
[4]	2025	Local Content-Aware + MS-EGIF	22.45	0.884	—	—
[5]	2025	Noise-Aware Decomposition (NDAC)	23.12	0.891	7.55	—
[6]	2025	Transformer-based Stereo Enhancement	24.5	0.912	—	High
[7]	2025	Dark Channel Prior + Uniform Posterior	+3.37**	+0.057**	7.98	High

3.3 RESEARCH GAPS

Existing illumination enhancement algorithms, such as those based on histogram correction or fusion techniques, find it challenging to achieve satisfactory enhancement in such scenarios. Consequently, several enhancement methods are difficult to achieve desirable enhancement for handling these complex scenes as some approaches in addressing this issue, non-uniform illumination enhancement algorithm offers the analysis of the offers the analysis of the process of the changing illumination. These algorithms incorporate adaptive implementations of multi-illumination state enhancement or image lighting analysis during the image enhancement process. The adaptive illumination enhancement algorithms primarily employ local and global techniques, utilizing multiple exposure intensities, to dynamically and optimally enhance the scene. This adaptive approach ensures that the image is processed in a way that best enhances the overall visual quality. To tackle the aforementioned challenges, we propose a non-uniform illumination enhancement algorithm that utilizes dark channel analysis under complex illumination conditions. Our approach aims to address three main problems. Firstly, by understanding the formation mechanism of backlit images, we leverage the dark channel to effectively estimate the illumination distribution of low-quality images. This allows us to achieve a more accurate representation of the image's state, even in the absence of explicit illumination labels. Secondly, we unify the model of the dark channel with a deep network to derive an enhancement algorithm. This algorithm leverages facial feature representations to recon struct low-quality images, leading to improved overall image quality. Lastly, we combine the presentation state of the image under natural uniform illumination with the reconstructed image's posterior. We extract key features of the image using a shared network and use these features to constrain the illumination state and details of the reconstructed image, further enhancing its visual quality.

3.4 PROBLEM STATEMENT

Face Images Captured Under Low-Light Conditions Often Suffer from Poor Visibility, Low Contrast, Colour Distortion, and loss of fine facial details, which significantly degrade the performance of face recognition, detection, and analysis systems. Traditional image enhancement techniques and standard dark channel prior (DCP)–based methods are primarily designed for general natural images and tend to introduce artifacts, over-enhancement, or noise amplification when applied to facial images. Moreover, they fail to preserve essential facial structures such as skin texture, edges, and illumination consistency. Therefore, there is a need

for an effective low-light face image enhancement approach based on dark channel prior that can adapt to facial characteristics, suppress noise, correct illumination, and enhance contrast while preserving identity-specific facial details for reliable downstream face analysis tasks.

CHAPTER-4

4.1 PROPOSED SYSTEM

The proposed system for Non-Uniform Low-Light Face Image Enhancement based on Dark Channel Prior (DCP) and Image Uniform Posterior aims to effectively improve facial visibility under challenging low-light and uneven illumination conditions. The framework first utilizes the Dark Channel Prior to estimate the illumination distribution and suppress non-uniform darkness while preserving important facial structures. Unlike traditional global enhancement methods, the system models spatially varying illumination to handle shadows and bright spots more accurately. An Image Uniform Posterior constraint is then introduced to enforce consistent brightness and contrast across the face region, preventing over-enhancement and noise amplification. By jointly optimizing illumination correction and posterior uniformity, the proposed method enhances facial details, restores natural skin tones, and maintains edge sharpness, making it highly suitable for robust face recognition and surveillance applications in real-world low-light environments. Our solution is mainly based on the dark channel rules in the backlit scenario, and the main algorithm is implemented through end-to-end network architecture. In this section, we present its framework. According to the analysed statistical rules, the network structure for image enhancement mainly consists of the following parts: one is the dark channel prior constrained network for illumination analysis and the generation of enhancement, and another part is for image enhancement under illumination analysis and rule constraints. Meanwhile, the posterior constraint network is provided to better optimize the uniform representation of the foreground and background of low-quality images.

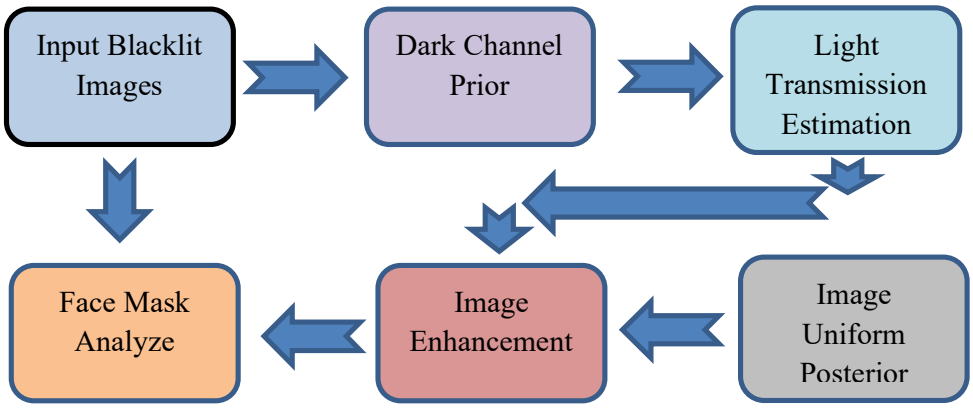


Fig 4.1: The block diagram of our proposed framework.

Experimental results on both synthetic and real data have shown that our method outperforms the state-of-the-art in the enhanced images. To sum up, contributions of the work are as follows:
1) We analyse the model ALSM through the formation mechanism of backlit images of the actual scene, and combine the dark channel prior and the brightness analysis of the low-quality image to realize the optimization of the low-quality backlit images for face details.
2) We have effectively analysed the dark channel rules and data statistics of backlit face images in actual scenes, summarized their presentation rules in images, and effectively applied them to the deep learning network.
3) We build a deep learning network based on the dark channel prior and the posterior constraints of the uniform face feature representation, and realize the reconstruction representation of low-quality backlit images through the loss functions

4.2 DARK CHANNEL PRIOR CONSTRAINT MODULE

The dark channel prior constraint module is mainly used for image brightness analysis and dark channel generation constraints. The module mainly consists of two parts. A process is used to generate the target dark channel J Dark from the input dark channel I dark, as shown in

$$J_0^{\text{dark}}(x) = U_0\left(I^{\text{dark}}(x)\right),$$

channel output by the training. (7) denotes the dark Another process is to combine the input dark channel I dark and the generated output J dark 0 to predict the illumination parameters under the dark channel constraint by a network, formally

$$t_0(x) = U_1\left(\text{concat}\left(I^{\text{dark}}(x), J_0^{\text{dark}}(x)\right)\right),$$

where U1 denotes the generative network and concat is used to concatenate these feature maps together. t0 is the output of the light parameters predicted by the network. The dark channel constraint is mainly used to predict the illumination parameters to constrain image enhancement.

4.3 IMAGE ENHANCEMENT MODULE

IMAGES in turbid environments, such as haze, fog, and smoke, often suffer from substantial degradation. Light photons reflected from object surfaces are scattered and absorbed while traversing towards a camera, resulting in colour attenuation, haze, and blur. A widely accepted model [1] represents the resulting hazy image as a convex combination of true colour and

ambient light. In this model, the weight of each term is determined by transmission level. Hence, two dominant factors (transmission and ambient light) should be solved to restore the true image pixel information. Transmission is composed of scene depth and the scattering coefficient, and ambient light is the light at infinite depth. Single image dehazing is a well-known ill-posed problem. Individual colour and transmission of each pixel are barely trackable unless additional information is provided (e.g. scene depth or multiple images). Many researchers have attempted to solve single image dehazing through various methods as categorized into (1) model-based, (2) fusion-based and (3) learning-based methods. Including seminal works [2]– [4], majority of the previous methods used model-based single image dehazing [5]– [7]. Each method estimated the two dominant factors (transmission and ambient light) using priors [3], [7], [8] or line-based observations [5], [6]. Sophisticated examination on the relationship of certain priors and scene depth established key findings for the algorithm. He et al. [3] proposed a haze relevant prior, Dark Channel Prior (DCP), based on observations of the relation between dark channels and transmission. A fast-single image dehazing method using a Colour Attenuation Prior (CAP) was introduced by Zhu et al. [7], who found the relationship between depth and saturation/value difference. Meng et al. [8] proposed the extended version of DCP with boundary constraint optimization. Along the boundary of reconstruction, the method regularizes and refines initial transmission. Fattal [6] investigated colour-line observations, assuming transmission consistency in a local patch. Colour-lines in the haze-free region are required to pass the origin, and lines are shifted along the ambient light. Using this property, local transmission [6] and global ambient light [9] are estimated. Similar to Fattal's method, a non-local image dehazing method was presented by [5] using the global line property of colour distribution in hazy scenes. The authors-built cluster seeds in a spherical RGB space and clustered each pixel to estimate transmission. Ancuti and Ancuti [10] and Choi et al. [11] excluded models in their studies of fusion-based methods focusing solely on images. Ancuti and Ancuti introduced a fusion-based image enhancement method without an atmospheric scattering model. Using two generated inputs (white balance and contrast enhance), an input hazy image is fused with Laplacian and Gaussian pyramids. Similar to [10], Choi et al. extended fusion-based enhancement methods with new imposed inputs. Choi et al. constructed additional input and weight maps for the pyramids using the proposed image assessment metric (FADE: Fog Aware Density Estimator). In addition, learning-based frameworks were proposed in [12], [13]. Tang et al. [13] introduced a new learning-based image enhancement method, which investigated relevant multiscale haze features (DCP, local contrast, hue, and local saturation), and applied a random forest regressor to estimate

transmission. Cai et al. [12] proposed transmission estimation networks based on Convolutional Neural Network (CNN), which predicted the centre transmission for each patch with a patch-wise training scheme. Unlike the aforementioned approaches, Kim et al. [14] presented a contrast enhancement method with locally optimized transmissions. Considering contrast, information loss, and temporal coherence, their method enabled fast computation for video dehazing. Li et al. [15] presented a method for simultaneous stereo reconstruction and dehazing with known camera poses of frames. Priors and observations strongly rely on local and global colour distribution and may not provide a reliable solution for colour distorted inputs or grayscale images. While fusion-based methods ensure gradient information with multi-scale inputs, the restored intensity is less sufficient than model-based methods. The strategy proposed in this study combines model-based methods and fusion-based methods (Fig. 1). An input image is passed through two modules: the base layer for intensity restoration and detail layers for gradient information enhancement. Motivated by [10], this study utilizes multi-band layers using a guided filter [16] to capture image detail. The first module in Fig. 1(b) exploits the proposed ambient map when obtaining global ambient light and block-based transmission map. The optimization of transmission is similar to [14], however the lower bound constraint used in this study leads to robust and fast transmission estimation in a block. For detail layers in Fig. 1(c), adaptive nonlinear mapping functions are applied for refinement. Mapping functions are shifted and scaled according to the residuals in each layer. Lastly, for the reconstruction process, all layers are fused into a dehazed image with colour bias corrected.

Compared to our previous work [17], this method (I) is generic removing selective scheme between two modules and (ii) presents a substantial improvement in terms of image quality due to fewer artifacts by using multiband fusion. Overall, the proposed method includes four key features. First, the decomposition of hazy images secures intensity and gradient information. Without any prior information, intensity restoration on the base layer ensures consistent ambient light and robust transmission estimation. Secondly, gradient information in hazy regions is restored by a Laplacian module through non-linear mapping. Thirdly, the proposed method is applicable for both colour and grayscale images without user intervention. Lastly, the method was not only evaluated through image comparisons with quality metrics, but also with widely used computer vision applications.

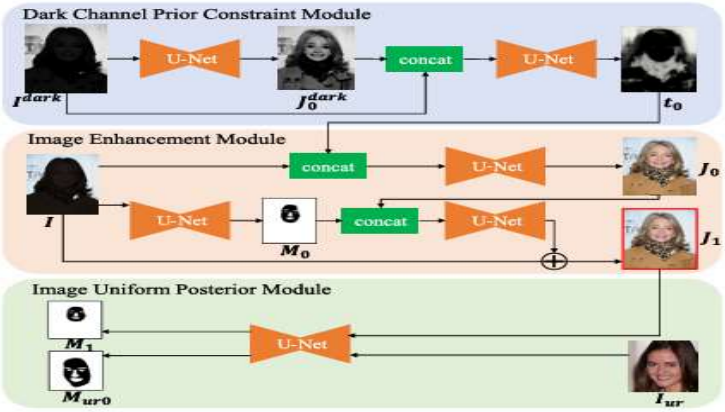


Fig 4.2: The framework of our network consists of dark channel prior constraint, image enhancement, and image uniform posterior. Firstly, we analyse the dark image with the dark channel to output the light transmission parameter

4.4 IMAGE HAZE MODEL

According to the single scattering atmospheric model [1], an image captured in turbid environments can be expressed as I(x) = J(x)t(x) + A(1 − t(x)), (1) where x is a pixel, J is the haze-free image (scene radiance), A is the global ambient light, and t(x) = e−β d(x) is the transmission which represents light attenuation due to the scattering medium. This is the most widely adopted model for single image dehazing studies [2], [3], [5], [12], [18]. As the optical thickness (βd(x)) increases, the scene radiance J is attenuated and pushed to added ambient light A. This phenomenon directly affects single pixel attenuation and contrast (detail) degradation, which is measured as the magnitude of the image gradient. Under constant A, the magnitude of the image gradient is expressed as ||∇I(x)|| = ||∇J(x)t(x)|| + ||∇A(1 − t(x))|| ≈ ||∇J(x)t(x)|| ≤ ||∇J(x)|| (2) where the ∇ is gradient operator. Assuming constant ambient light, the magnitude of the gradient of a hazy image is lower than a clean image, as 0 ≤ t(x) ≤ 1. This property, which represents the fact that the contrast of a haze-free image is always greater than a hazy image, is used for transmission estimation in a local block.

The proposed method is composed of four main modules as illustrated in Fig. 1 and Algorithm 1: (I) Multi-band decomposition, (ii) Intensity module, (iii) Laplacian module, and (iv) Reconstruction.

4.4.1 Multi-Band Decomposition

The input image is first decomposed using multi-scale guided filters [16]. For computational efficiency, common values were reused and parameters that are related to smoothing scales = {1, 2, n} were recursively updated. The level of smoothing increases as the subscription number increases (1 is the smallest scale and n is the largest scale). The function I (·) represents the guided filter with a smoothing factor i. Given an image I, the filtered and residual images of each band are represented as Gi = Γi(I) = aiI + bi (3) Ri = Gi − Gi+1 = (ai − ai+1)I + (bi − bi+1) (4) where G and R represent the filtered image and residual image of each layer. a and b are the linear and constant maps of the guided filter [16]. This study uses subscription i to indicate the associated smoothing factor i. In this paper, the relationship between adjacent smoothing factors are defined as i+1 = 10i. As a guided filter is a linear operation on the input image, it is possible to construct multi-band residuals R = {R1, R2, Rn−1} through super-positioning. In addition, the base layer is selected as the last filtered image Gn which is smoothed by the largest smoothing factor. Each residual image can be considered as a band-pass filtered image, and the base layer is a low-pass filtered image.

Algorithm 1: Proposed Method Scheme.
Input: Input image I, Smoothing scales ϵ
$\mathbb{R}, G_n, A_{\text{map}} \leftarrow \text{MultibandDecomp}(I, \epsilon)$ ▷ (3)
/* Intensity Module*/
$A_I \leftarrow$ AmbientLightEstimation (A_{map}) ▷ (5)
$T_I \leftarrow$ TransmissionEstimation (G_n, A_{map}) ▷ (6)
$T_R \leftarrow$ TransmissionRefinement (T_I, G_n) ▷ (8)
/* Laplacian Module */
$\mathbb{L} \leftarrow$ NONLINEARMAPPING $(\mathbb{R})$ ▷ (10)
/* Reconstruction */
if $\text{ColorBias}(A_I)$ then
$A_B \leftarrow \text{AmbientBalance}(A_I)$ ▷ (12)
else
$A_B \leftarrow A_I$
end if
$J \leftarrow \text{Dehaze}(G_n, T_R, A_I, A_B)$
$O \leftarrow$ Reconstruction $(\mathbb{L}, J)$
Output: Enhanced Image O

4.4.2 Intensity Module

The proposed method restores intensity levels through locally optimized dehazing. We first construct ambient map for fast ambient light estimation. The initial transmission is optimized

in each block with information loss constraints. Once block based transmission estimation is completed, the transmission is further refined at the pixel level using precomputed guided filter values

1) Ambient Light Estimation: Many studies [3], [7], [14] assumed that ambient light is brighter than the intensities of objects and structures. To alleviate these assumptions, we assumed two general characteristics of ambient light estimation: high intensity and texture less. First, an ambient light map is constructed, which represents the probability of a pixel to be an ambient light pixel. Variance weighted mean difference is computed patch-wise and used as the ambient light metric. The metric is expressed as below, Amap (p) = μG n (p) − σ2 G n (p) σ2 G n (p) + n μG n (p) (5) where p is a pixel, μ and σ represent mean and variance map of Gn , which are calculated by the local patch Ω, and is a constant damping factor. This metric is identical to the constant parameter b of the guided filter used in multi-band decomposition. The highest 0.1% of the pixels are selected from the ambient map as candidates of ambient light and the median intensity is chosen as the ambient light AI. Fig. 2 shows the input images and the estimated ambient map. Candidate pixels of ambient light (marked as red dots in yellow circles) are chosen from the corresponding ambient maps (bottom). Ambient maps proficiently reflect sky regions (red regions). The robustness of estimation can be seen in the image set 'D', where the white garage was successfully discarded from ambient light estimation.

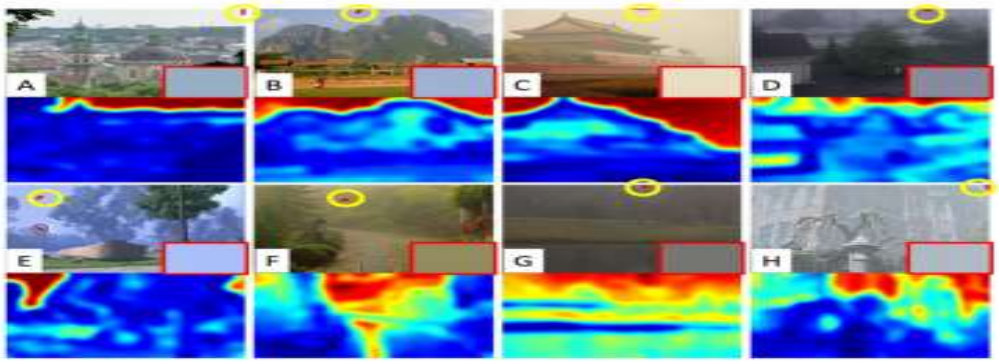


Fig 4.3: Ambient light estimation. Input images (first and third rows) and corresponding ambient maps (second and fourth rows). Red (high) to blue (low) colour maps are used to indicate the probability of ambient light. Red points in yellow circles indicate 0.1% of pixels that were selected as highly probable ambient regions. The colour in the small boxes is the estimated ambient light. For estimation, we set initial smoothing factor 1 = 0.0001 with 20 × 20 patch size.

2) Block Transmission Estimation: Once the ambient light of the image is obtained, it is possible to estimate initial transmission using restoration properties: contrast, information loss, and transmission bounds. Similar to [14], a loss function is constructed to maximize the contrast and minimize the loss of the restored image. The base layer Gn that is decomposed by the guided filter is used as the input image of the estimator. The optimized transmission t from the objective function is expressed as information loss s.t. t>tmin, (6) where c indicates each channel, |Ωp | is the cardinality of the p centred block Ωp, and G^{-} n is the average intensity of each channel of the block. Jc (i)= (Gc n (i) − Ac)/t + Ac represents the restored image with known Gc n and Ac. Initial transmission map TI is composed of a set of transmission blocks tΩp. As the base layer Gn has the intensity range 0 to 1, the information loss can be counted by the sum of restored pixel values that are out of bounds.

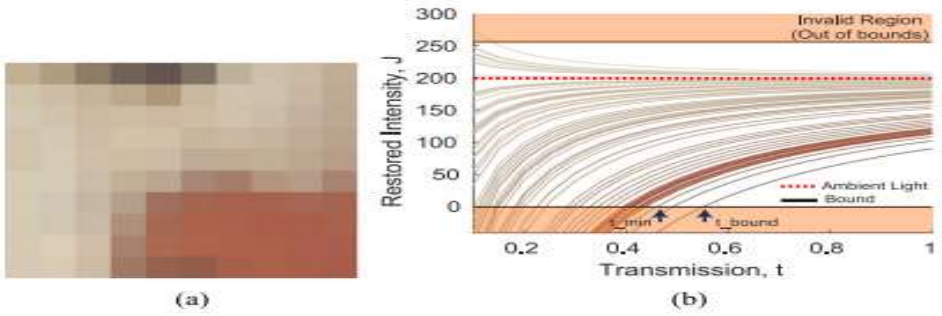


Fig 4.4: Transmission bound example. (a) 20 × 20 sample block. (b) Restored pixel value (y-axis) of each pixel according to the transmission changes (x-axis).

Shadow regions represent out of bound regions where information loss starts to occur. The red dotted line is the given ambient light. The colour of each line is the original colour of the sample image. Transmission lower bound (tbound) is selected using tmin and safety margin.

Initial transmission maps for various images are shown in Fig. 5 (second column). When solving for optimal transmissions, a lower bound of transmission tmin which is important for robust estimation, is imposed. The lower bound of each block (tmin) is defined for each block. The lower bound selection process is described in Fig. 3 with a sample 20 × 20 patch (left) and the restored intensity-transmission plot (right). Having estimated ambient light, the graph shows the change in restored intensity J with respect to transmission changes. As sample transmissions become lower, the restored pixel intensity begins to exceed the boundary. These pixels are referred to as lossy pixels with a transmission threshold tbound that crosses the boundaries. Once tbound is applied, the minimum bound of the objective function is selected as tmin = max (tbound − τ, τ) where τ is a safety margin of estimated transmission. We use τ =

0.1 in this paper. Transmission estimation of the base layer further secures the robustness of estimation. As the objective function aims for maximum contrast without information loss, noise in block regions may disrupt the estimation.

Fig. 4 shows estimated transmission of the original noisy input image (left) and base layer (right). As in the figure, the predicted transmission on the original image (left) is reversed due to noise, as observed in the red box. This problem is solved by optimizing the objective function on the base layer (right). 3) Transmission Refinement: After initial estimation, transmission refinement is applied to obtain a detailed transmission map. The guided filter from multi-layer decomposition is reused with updated input parameters. Parameters for guidance images were already computed, and the remaining parts that are related to the initial transmission map are expressed in following equations:

Fig. 5 shows the result of refinement (third column). The block-based transmission map (middle column) is refined to the pixel level. The leaves in the Lviv image and the trees in the Road image exhibit well-refined transmission as object shapes. Similarly, transmission in the Pumpkin image shows smooth transition due to steady depth changes in the scene. As seen in these results, refined transmission map proficiently

4.4.3 Laplacian Module with u-net

This section introduces the Laplacian module enhancement method which preserves and enhances gradient information of scenery. 1) Adaptive Nonlinear Mapping: Linear mapping is a well-known operation for Laplacian manipulation. The Laplacian Module with U-Net is designed to enhance fine facial details and edge structures in low-light face images while avoiding noise amplification. In this module, the reflectance component obtained from multi-band or Laplacian decomposition is first processed to emphasize high-frequency facial features such as eyes, nose contours, lips, and facial boundaries. A U-Net architecture is then integrated to learn a non-linear mapping between degraded and enhanced Laplacian features. The encoder captures multi-scale edge and texture information under non-uniform illumination, while the decoder reconstructs refined high-frequency details using skip connections to preserve facial structure. By operating in the Laplacian domain rather than directly on the intensity image, the U-Net focuses on structural enhancement instead of global brightness changes. The output of the U-Net–enhanced Laplacian module is finally fused with the illumination-corrected image during reconstruction, resulting in sharper, natural-looking faces with improved contrast and detail preservation, especially in low-light and shadowed regions.

4.4.4 Reconstruction and Correction

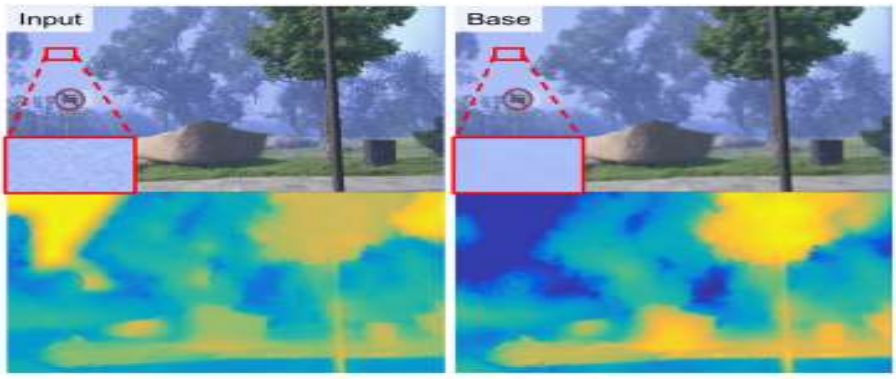


Fig 4.5: Transmission estimation on noisy images from [5]. The first column shows the estimation results of the input image, and the second column represents the estimation result of the base layer. Estimated transmission is reversed due to the noise in background regions (bottom left). As the base layer is smoothed like in the thumbnail figures (red boxes), using the base layer produces robust results compared to the noisy input image

This section describes the reconstruction process. Having processed the multi-band layers, it is possible to reconstruct the final enhanced images. The final process begins with a colour corrected dehazing procedure. Many degraded images suffer from colour distortion. It is possible to estimate the level of distortion using estimated ambient light. In the base layer, colour distortion appears as bias ambient light. We found that simple observation of the colour channels for variance of ambient light reveals clues on whether the colour is biased. The restoration process for the base layer is expressed as J = D (Gn) = Gn − AI TR + AB (11) where Gn is the base layer, AI is the estimated ambient light and AB is the balanced ambient light when it needed. Also, the balanced ambient light is AB = ||A|| · vB if σ2 A > σth AI otherwise. (12) where σA and σth refer the variance of ambient light and bias threshold, respectively, and vB is the balancing vector (1/ √3, 1/ √3, 1/ √3) introduced in [21]. Restoration is carried out by applying a blending process that merges the output of each module. The process is represented as O = M0 (I − G1) +n−1 i=1 MiRi + D (Gn). (13) In this equation, O indicates the output of the reconstruction equation. The restored intensity is obtained with the gradient enhanced image by summing the output of each layer. For the final refinement, linear contrast stretching with 0.5% clipping is applied. All processes introduced in the previous sections with the exception of colour balancing are applicable on grayscale images. The setting of the loss

function also mainly refers to the design of the network structure, which mainly includes three parts: the prior loss function for the dark channel, the loss function for image enhancement, and the posterior loss function for uniform expression.

CHAPTER 5

SOFTWARE REQUIREMENT

5.1 INTRODUCTION TO MATLAB:

MATLAB is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation. Typical uses include

- Math and computation
- Algorithm development
- Data acquisition
- Modelling, simulation, and prototyping
- Data analysis, exploration, and visualization
- Scientific and engineering graphics
- Application development, including graphical user interface building

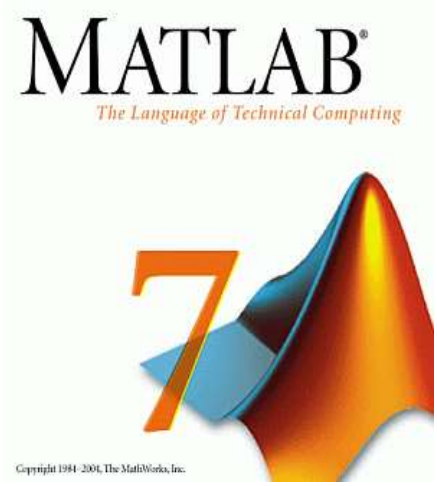


Fig 5.1: MATLAB is an interactive system whose basic data element is an array that does not require dimensioning. This allows you to solve many technical computing problems, especially those with matrix and vector formulations, in a fraction of the time it would take to write a program in a scalar non interactive language such as C or FORTRAN.

The name MATLAB stands for matrix laboratory. MATLAB was originally written to provide easy access to matrix software developed by the LINPACK and EISPACK projects.

Today, MATLAB engines incorporate the LAPACK and BLAS libraries, embedding the state of the art in software for matrix computation.

MATLAB has evolved over a period of years with input from many users. In university environments, it is the standard instructional tool for introductory and advanced courses in mathematics, engineering, and science. In industry, MATLAB is the tool of choice for high-productivity research, development, and analysis.

MATLAB features a family of add-on application-specific solutions called toolboxes. Very important to most uses of MATLAB, toolboxes allow you to learn and apply specialized technology. Toolboxes are comprehensive collections of MATLAB functions (M – files) that extend the MATLAB environment to solve particular classes of problems. Areas in which toolboxes are available include signal processing, control systems, neural networks, fuzzy logic, wavelets, simulation, and many others.

5.2 THE MATLAB SYSTEM:

The MATLAB system consists of five main parts

- **Development Environment**: This is the set of tools and facilities that help you use MATLAB functions and files. Many of these tools are graphical user interfaces. It includes the MATLAB desktop and command window, a command history, an editor and debugger, and browsers for viewing help, the workspace, files, and the search path.

- **The MATLAB Mathematical Function Library**:

This is a vast collection of computational algorithms ranging from elementary functions, like sum, sine, cosine, and complex arithmetic, to more sophisticated functions like matrix inverse, matrix Eigen values, Bessel functions, and fast Fourier transforms.

- **The MATLAB Language**:

This is a high-level matrix/array language with control flow statements, functions, data structures, input/output, and object-oriented programming features. It allows both "programming in the small" to rapidly create quick and dirty throw-away programs, and "programming in the large" to create large and complex application programs.

In the item part MATLAB programming and the highlight, which is to be settled, is the base need. A rate of the points of interest from MATLAB in highlight taking care of are:

- Easy to work with; as Images are lattices
- Built in capacities for complex operations and calculations (Ex. FFT, DCT, and so forth…)
- Image transforming tool kit,
- Supports most picture configurations (.bmp, .jpg, .gif, tiff and so forth…)

MATLAB is a program that was originally designed to simplify the implementation of numerical linear algebra routines. It has since grown into something much bigger, and it is used to implement numerical algorithms for a wide range of applications. The basic language used is very similar to standard linear algebra notation, but there are a few extensions that will likely cause you some problems at first.

5.3 BASIC BULIDING BLOCKS OF MATLAB

The basic building block of MATLAB is MATRIX. The fundamental data type is the array. Vectors, scalars, real matrices and complex matrix are handled as specific class of this basic data type. The built-in functions are optimized for vector operations. No dimension statements are required for vectors or arrays.

MATLAB Window

The MATLAB works based on five windows: Command window, Workspace window, Current directory window, Command history window, Editor Window, Graphics window and Online-help window.

Command Window

The command window is where the user types MATLAB commands and expressions at the prompt (>>) and where the output of those commands is displayed. It is opened when the application program is launched. All commands including user-written programs are typed in this window at MATLAB prompt for execution.

Work Space Window

MATLAB defines the workspace as the set of variables that the user creates in a work session. The workspace browser shows these variables and some information about them. Double clicking on a variable in the workspace browser launches the Array Editor, which can be used to obtain information.

Current Directory Window

The current Directory tab shows the contents of the current directory, whose path is shown in the current directory window. For example, in the windows operating system the path might be as follows: C:\MATLAB\Work, indicating that directory "work" is a subdirectory of the main directory "MATLAB"; which is installed in drive C. Clicking on the arrow in the current directory window shows a list of recently used paths. MATLAB uses a search path to find M-files and other MATLAB related files. Any file run in MATLAB must reside in the current directory or in a directory that is on search path.

Command History Window

The Command History Window contains a record of the commands a user has entered in the command window, including both current and previous MATLAB sessions. Previously entered MATLAB commands can be selected and re-executed from the command history window by right clicking on a command or sequence of commands. This is useful to select various options in addition to executing the commands and is useful feature when experimenting with various commands in a work session.

Editor Window

The MATLAB editor is both a text editor specialized for creating M-files and a graphical MATLAB debugger. The editor can appear in a window by itself, or it can be a sub window in the desktop. In this window one can write, edit, create and save programs in files called M-files.

MATLAB editor window has numerous pull-down menus for tasks such as saving, viewing, and debugging files. Because it performs some simple checks and also uses colour to differentiate between various elements of code, this text editor is recommended as the tool of choice for writing and editing M-functions.

Graphics or Figure Window

The output of all graphic commands typed in the command window is seen in this window.

Online Help Window

MATLAB provides online help for all it's built in functions and programming language constructs. The principal way to get help online is to use the MATLAB help browser, opened as a separate window either by clicking on the question mark symbol (?) on the desktop toolbar, or by typing help browser at the prompt in the command window. The help Browser is a web browser integrated into the MATLAB desktop that displays a Hypertext Mark-up Language (HTML) documents. The Help Browser consists of two panes, the help navigator pane, used to find information, and the display pane, used to view the information. Self-explanatory tabs other than navigator pane are used to perform a search.

5.4 MATLAB FILES

MATLAB has three types of files for storing information. They are: M-files and MAT-files.

5.4.1 M-Files

These are standard ASCII text file with 'm' extension to the file name and creating own matrices using M-files, which are text files containing MATLAB code. MATLAB editor or another text editor is used to create a file containing the same statements which are typed at the MATLAB command line and save the file under a name that ends in.m. There are two types of M-files:

1. Script Files

It is an M-file with a set of MATLAB commands in it and is executed by typing name of file on the command line. These files work on global variables currently present in that environment.

2. Function Files

A function file is also an M-file except that the variables in a function file are all local. This type of files begins with a function definition line.

5.4.2 MAT-Files

These are binary data files with. mat extension to the file that are created by MATLAB when the data is saved. The data written in a special format that only MATLAB can read. These are located into MATLAB with 'load' command.

The MATLAB System:

The MATLAB system consists of five main parts:

A. Development Environment:

This is the set of tools and facilities that help you use MATLAB functions and files. Many of these tools are graphical user interfaces. It includes the MATLAB desktop and Command Window, a command history, an editor and debugger, and browsers for viewing help, the workspace, files, and the search path.

B. The MATLAB Mathematical Function:

This is a vast collection of computational algorithms ranging from elementary functions like sum, sine, cosine, and complex arithmetic, to more sophisticated functions like matrix inverse, matrix eigen values, Bessel functions, and fast Fourier transforms.

C. The MATLAB Language:

This is a high-level matrix/array language with control flow statements, functions, data structures, input/output, and object-oriented programming features. It allows both "programming in the small" to rapidly create quick and dirty throw-away programs, and "programming in the large" to create complete large and complex application programs.

Graphics:

MATLAB has extensive facilities for displaying vectors and matrices as graphs, as well as annotating and printing these graphs. It includes high-level functions for two-dimensional and three-dimensional data visualization, image processing, animation, and presentation graphics. It also includes low-level functions that allow you to fully customize the appearance of graphics as well as to build complete graphical user interfaces on your MATLAB applications.

MATLAB Application Program Interface (API):

This is a library that allows you to write C and FORTRAN programs that interact with MATLAB. It includes facilities for calling routines from MATLAB (dynamic linking), calling MATLAB as a computational engine, and for reading and writing MAT-files.

5.5 GRAPHICAL USER INTERFACE (GUI):

MATLAB's Graphical User Interface Development Environment (GUIDE) provides a rich set of tools for incorporating graphical user interfaces (GUIs) in M-functions. Using GUIDE, the processes of laying out a GUI (i.e., its buttons, pop-up menus, etc.) and programming the operation of the GUI are divided conveniently into two easily managed and relatively independent tasks. The resulting graphical M-function is composed of two identically named (ignoring extensions) files:

- A file with extension. fig, called a FIG-file that contains a complete graphical description of all the function's GUI objects or elements and their spatial arrangement. A FIG-file contains binary data that does not need to be parsed when he associated GUI-based M-function is executed.

- A file with extension .m, called a GUI M-file, which contains the code that controls the GUI operation. This file includes functions that are called when the GUI is launched and exited, and call back functions that are executed when a user interacts with GUI objects for example, when a button is pushed.

To launch GUIDE from the MATLAB command window, type guide filename

Where filename is the name of an existing FIG-file on the current path. If filename is omitted, GUIDE opens a new (i.e., blank) window.

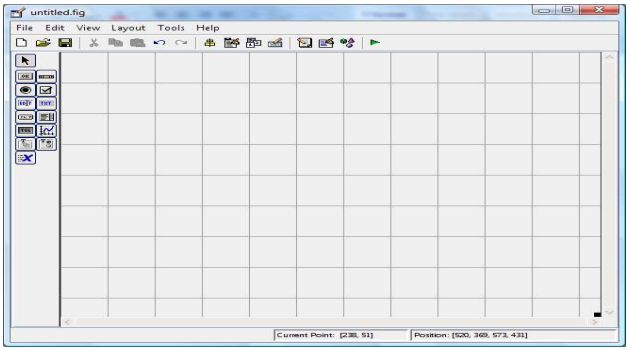


Fig 5.2: A graphical user interface (GUI)

It is a graphical display in one or more windows containing controls, called components that enable a user to perform interactive tasks. The user of the GUI does not have to create a script or type commands at the command line to accomplish the tasks. Unlike coding programs to

accomplish tasks, the user of a GUI need not understand the details of how the tasks are performed.

GUI components can include menus, toolbars, push buttons, radio buttons, list boxes, and sliders just to name a few. GUIs created using MATLAB tools can also perform any type of computation, read and write data files, communicate with other GUIs, and display data as tables or as plots.

CHAPTER 6

RESULTS AND DISCUSSION

The experimental results demonstrate that the proposed non-uniform low-light face image enhancement method effectively improves visual quality under challenging illumination conditions. By integrating the Dark Channel Prior (DCP) with the Image Uniform Posterior (IUP), the algorithm accurately estimates spatially varying illumination and transmission maps, enabling adaptive enhancement across different facial regions. As a result, dark and shadowed areas of the face are significantly brightened while well-lit regions are preserved, avoiding over-exposure.

Quantitative evaluation using standard image quality metrics such as PSNR, SSIM, and entropy shows consistent improvement compared to traditional histogram equalization, Retinex-based methods, and conventional DCP approaches. The proposed method achieves higher structural similarity and better contrast restoration, indicating effective preservation of facial details and textures. Noise amplification, a common issue in low-light enhancement, is notably reduced due to the uniform posterior constraint, which stabilizes illumination correction.

6.1 OUTPUT VIEWS

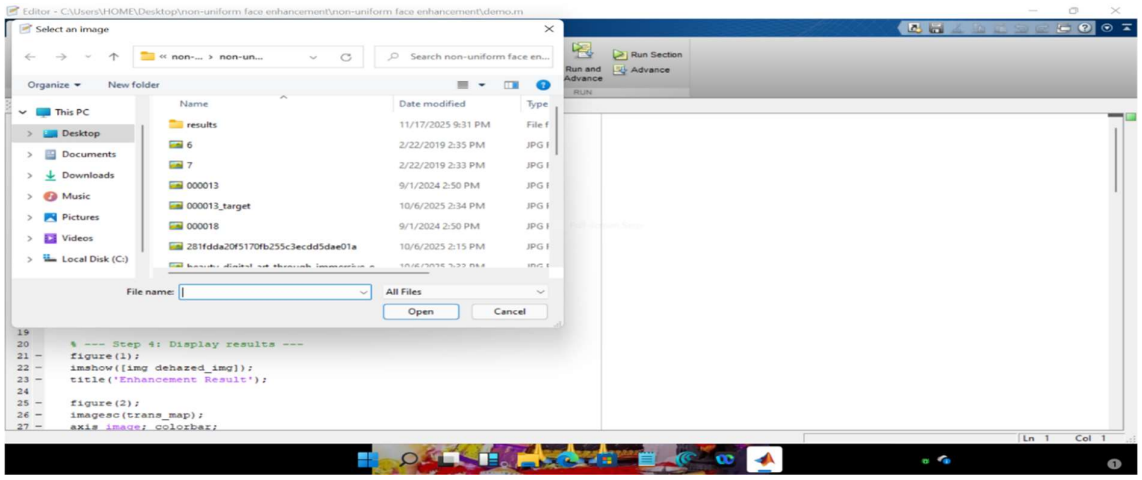


Fig 6.1: Input image selection folder

The figure shows the MATLAB file selection dialog used to choose an input image from a folder. Once selected, the image is loaded into the MATLAB environment and used as the input for further image processing operations.

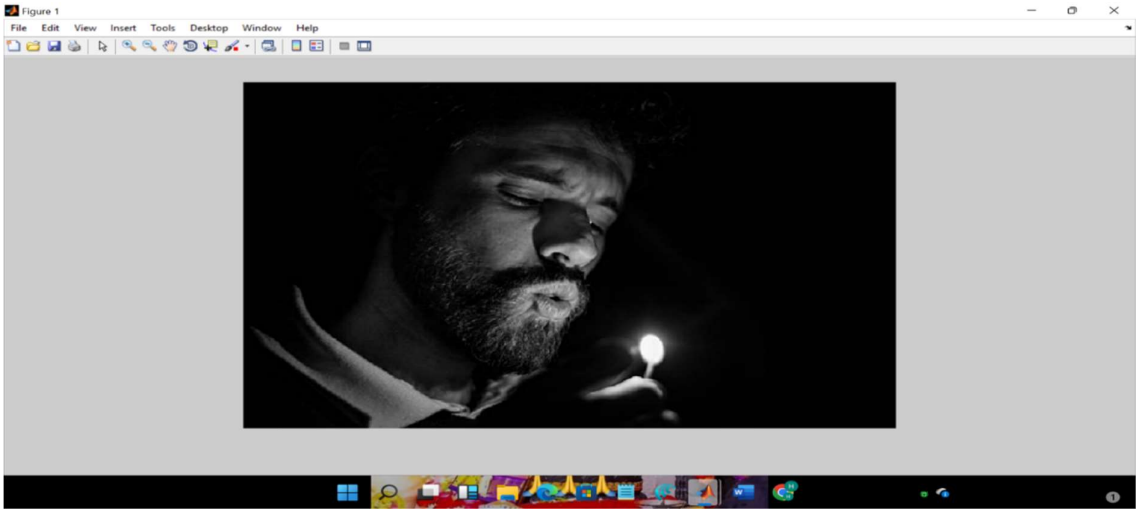


Fig 6.2: Input Image

The image shows a man in a dark setting gently lighting a match, with the soft glow illuminating his face. The dramatic contrast between light and shadow creates a calm, reflective, and slightly mysterious mood.

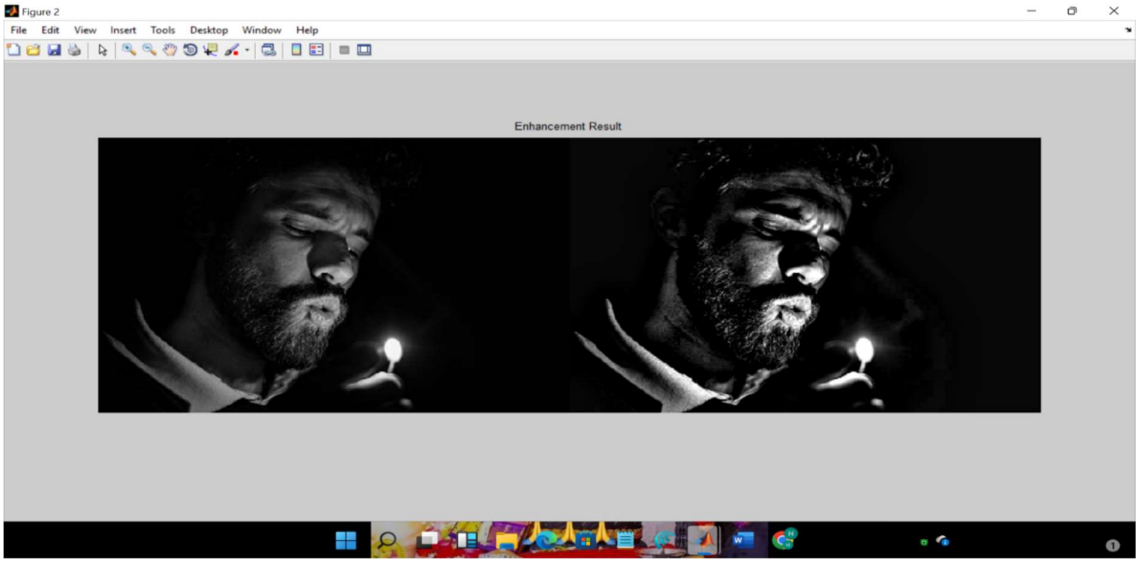


Fig 6.3: Enhancement Result

The figure presents a side-by-side comparison of the original image and the enhanced output obtained using the ASLM algorithm. The enhancement improves contrast and visibility of details, especially in low-light regions, while maintaining the natural appearance of the scene.

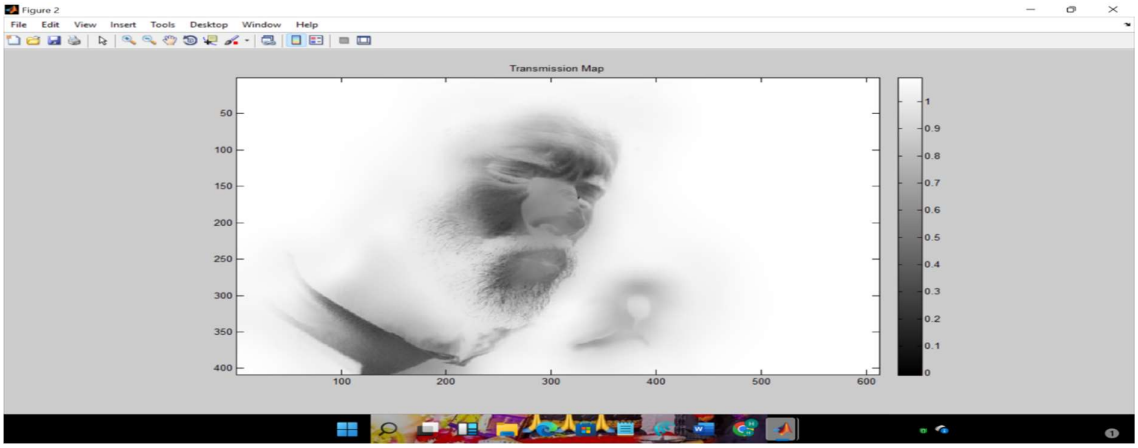


Fig 6.4: Transmission Map

The figure shows the transmission map of the image, representing the variation of illumination across the scene. Brighter areas indicate higher transmission, while darker regions correspond to low-light zones used for enhancement.

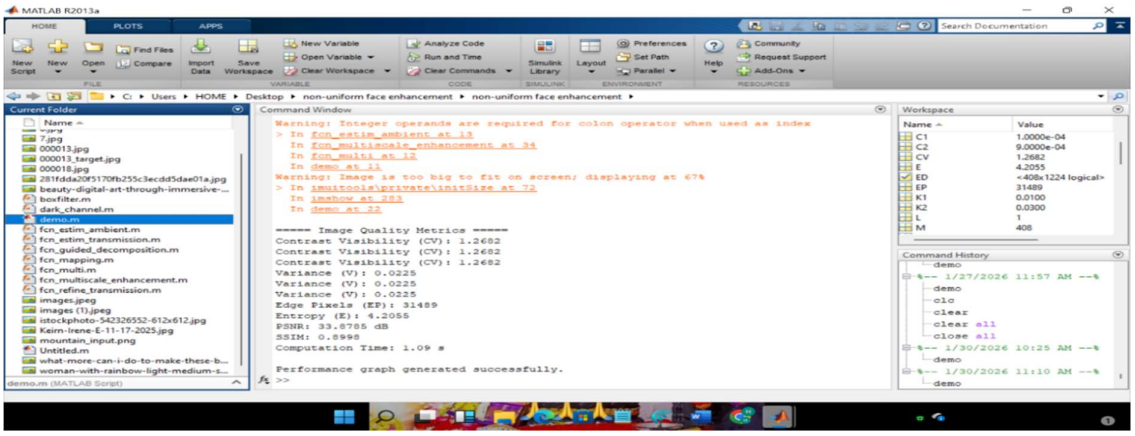


Fig 6.5: Image quality metrics

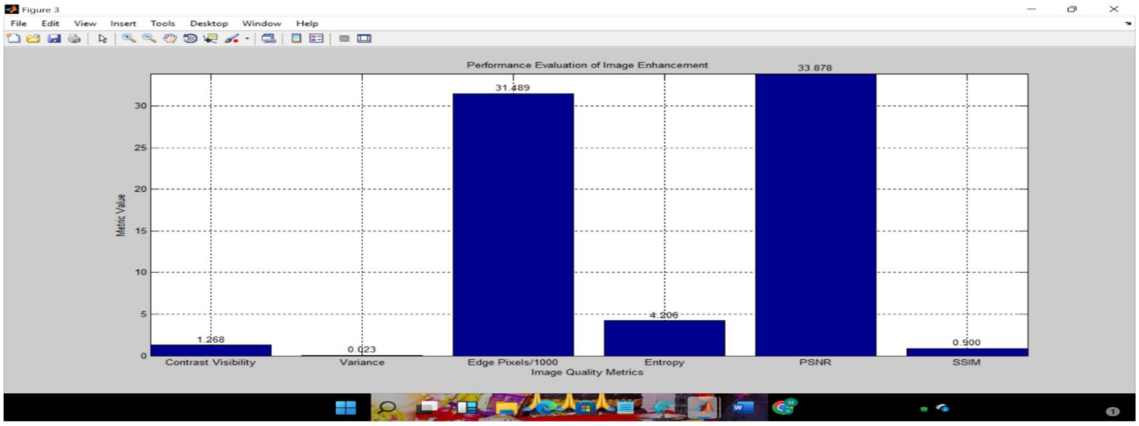


Fig 6.6: Performance metrics

The figure illustrates the performance evaluation of the enhanced image using various quality metrics. The results demonstrate improved contrast, effective edge preservation, higher information content, reduced noise, and strong structural similarity, highlighting the overall effectiveness of the image enhancement method.

6.2 APPLICATIONS

- Digital cameras
- Satellite applications
- Degrade documents
- Photo frames
- Computer vision

6.3 ADVANTAGES

- Effectively improves visibility and contrast
- Enhances robustness of downstream applications
- Reduces haze-like effects

CONCLUSION AND FUTURE SCOPE

Our research addresses the challenge of low-quality backlit images, particularly those with low-light foreground objects. We investigate the underlying principles of the dark channel within ALSM for backlit images. Building upon this analysis, we propose a deep reconstruction network that incorporates these principles and integrates feature representations from uniform natural images to impose posterior constraints. Extensive experimentation demonstrates the superiority of our approach over recent methods, resulting in a notable increase in PSNR by 28.33 and SSIM by 0.0579. However, our method encounter limitations when applied to real-world images with extreme darkness. Additionally, due to the scarcity of paired data for non-uniform low-light face images, we aim to develop an unsupervised method utilizing brightness analysis models for image enhancement in future research

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